

Isaac Kaplan

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isaac-kaplan.com

Education

Carnegie Mellon University | Pittsburgh, PA

Expected Graduation Date: May 2026

Bachelor's of Science: Mechanical Engineering

Bachelor's of Science: Computer Science

GPA: 3.61/4.0

Experience

Research Experience Undergraduate (REU) | Kent, Ohio | Summer 2025

Kent State University (Cognitive AI and Robotics lab)

- Programmed a natural language command parser and runner integrated into a ROS system.
- Collected data and fine-tuned GPT-4o using OpenAI fine-tuning.

Student Undergraduate Research Assistant (SURA) | Pittsburgh, PA | Summer 2024

Carnegie Mellon University (Legged Systems Group)

- Developed an MLP Neural Network to make path predictions for a robotic prosthetic leg.
- Modified and improved the control code for the leg to run with the AI system.

Software-Development-Intern | Boulder, CO | Summer 2023

Software-Development-Experts

- Developed data processing software to scrape from an API and process it to extract and reformat information.
 - Set up an auto-deploying website's infrastructure using Jenkins, Docker, and Kubernetes.
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Projects

Carnegie Mellon Rocket Command | Fall 2022 – Fall 2023

- Programmed a Raspberry Pi to receive and decode radio broadcasts, then move servos and take photos based on the contents of the transmission.

Space Hanger Project | Mechanical Design Fall 2023

- Designed a member out of acrylic to withstand a 50 lb load and thermal expansion and minimized mass to 3.7 grams.
 - Iterated design and testing using SolidWorks simulation and experimentation.
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Leadership Experience

Teaching Assistant for Fundamentals of Programming (15-112) | Fall 2023 - Present

Carnegie Mellon University

- Taught recitation twice a week as well as mentoring 11 students through making a ~1000 line project from scratch.

Resident Advisor | Fall 2023

Carnegie Mellon University

- Planned large events for my housing community and used conflict resolution and mediation skills to resolve disagreements between residents.
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Skills

Mechanical Engineer Skills: CAD, Matlab, Excel, Laser cutting, 3D printing, FEA, Ansys Discovery, Arduino, Raspberry Pi

Computer Science Skills: C, OCaml, Python, SML, VBA, Pytorch, Computer Systems, OOP, Compiler Design